

Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School

Week 1

MONDAY

Day 1 - Procedures, Lab Safety & Design Notebook

Aug 10

OBJECTIVE

Students will set up their Year-2 design notebook (refresher), learn the lab-safety non-negotiables, and sign into Google Classroom.

TEKS

§127.813(c)(1) safety, professional standards & responsible tool use

ELPS / EB SUPPORT

Safety rules shown with icons + demo (VC). EB students paired with a bilingual peer for sign-in (PN, CD).

AGENDA • 45 MIN

6' Welcome back + how the shop runs (Year 2)

16' Lab & shop safety — the non-negotiables

15' Notebook refresh + set up Year-2 sections

8' Sign into Google Classroom + exit ticket

SLIDES / ACTIVITY

Welcome to Manufacturing

NOTEBOOK

Set up cover, table of contents, numbered/dated pages. First entry: one thing you want to design and build this year.

TURN IN

Design Notebook - Year 2 Setup (refresher), Lab Safety - Agreement & Check

TUESDAY

Day 2 - Manufacturing Systems

Aug 11

OBJECTIVE

Students will explain how manufacturing systems organize inputs, processes, and outputs, and classify job, batch, and continuous production.

TEKS

§127.813 (draft, verify strand) manufacturing systems - inputs, processes, outputs & production types

ELPS / EB SUPPORT

Vocabulary handout with pictures (VC, PV). Scenario sort done in pairs so EB students talk it through (PN).

AGENDA • 45 MIN

5' Bell-work: how is a car different from a wedding cake to make?

12' Watch Toyota + systems notes

10' Job / Batch / Continuous mini-lesson

13' Vocab match + scenario sort

5' Exit ticket

SLIDES / ACTIVITY

How Things Get Made at Scale

NOTEBOOK

Draw the input -> process -> output diagram. List one product for each production type (job, batch, continuous).

TURN IN

Manufacturing Systems - Vocabulary & Scenario Sort

WEDNESDAY

Day 3 - Planning & Logistics

Aug 12

OBJECTIVE

Students will use planning and logistics vocabulary and map the workflow of a simple product.

TEKS

§127.813 (draft, verify strand) production planning, logistics & workflow

ELPS / EB SUPPORT

Workflow diagram is visual (VC). Word bank provided; pairs build the diagram together (PN, CD).

AGENDA • 45 MIN

5' Bell-work: what is a bottleneck?

12' Planning & logistics mini-lesson

8' Vocab (JIT, supply chain, cycle time)

15' Workflow-diagram task for a real product

5' Exit

SLIDES / ACTIVITY

Getting the Right Parts to the Right Place

NOTEBOOK

Draw a workflow diagram for a simple product (e.g., a sandwich shop). Label one bottleneck.

TURN IN

Logistics Vocabulary + Workflow Diagram

THURSDAY**Day 4 - The Industry, Careers & Safety First**

Aug 13

OBJECTIVE

Students will describe the modern manufacturing industry, explore a related career, and read a basic safety data sheet.

TEKS

§127.813 (draft, verify strand) industry overview, careers & workplace safety

ELPS / EB SUPPORT

Career sheet uses a template with sentence stems (CD, RS). Factory video builds background knowledge (VC).

AGENDA • 45 MIN

5' Bell-work: name a product made near El Paso

12' Industry + careers + watch factory tour

15' Career one-pager (pick a manufacturing career)

8' Safety-data-sheet mini task

5' Share + exit

SLIDES / ACTIVITY

The Manufacturing Industry Today

NOTEBOOK

Note one manufacturing career: what they do, pay range, and education needed.

TURN IN

Manufacturing Career - One-Pager

FRIDAY**Day 5 - Design Process -> Your Product Idea**

Aug 14

OBJECTIVE

Students will apply the engineering design process to define a product problem and brainstorm solutions they could later build in Fusion.

TEKS

§127.813 (draft, verify strand) engineering design process applied to a manufacturable product

ELPS / EB SUPPORT

Product-idea sheet is heavily scaffolded (CD). Brainstorm in pairs; drawings welcome instead of words (VC, PN).

AGENDA • 45 MIN

5' Bell-work: a product that annoys you - why?

10' Design process for manufacturing

22' Develop Your Own Product Idea (Ask -> Research -> Brainstorm)

8' Share ideas + exit

SLIDES / ACTIVITY

Design Something Worth Making

NOTEBOOK

Glue in the product-idea sheet. Sketch your top idea and list what it's made of.

TURN IN

Develop Your Own Product Idea